

BridgeFinder: Finding People Across Structural Holes

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CS 470 project proposal, University of Illinois Urbana-Champaign

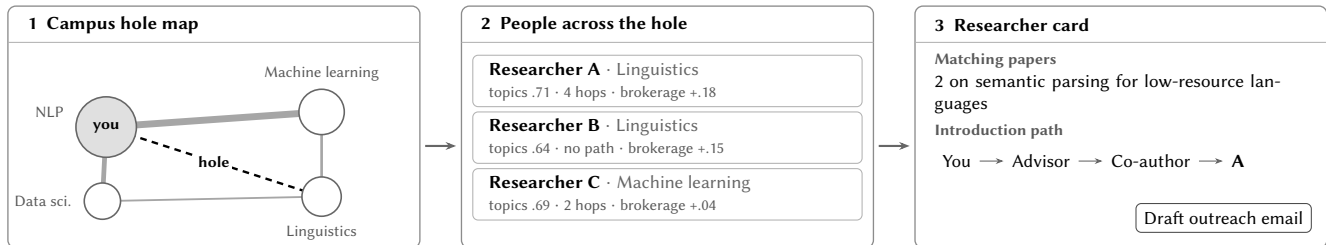


Figure 1: Core flow. (1) UIUC co-authorship communities; dashed = a structural hole (similar topics, almost no joint papers). (2) Researchers across the hole, ranked. (3) Their matching papers and an introduction path.

1 OBJECTIVE

Students seeking a research mentor, and researchers seeking collaborators, search through people they already know: an advisor's contacts, their lab, their department. That returns people who are already close. Yet new information travels through weak ties [5], and people who broker between groups that don't talk have better ideas [4]. At UIUC ([N] researchers, [M] units), two groups can work on the same problem for years without a joint paper. **BridgeFinder** finds these gaps in the co-authorship graph and names the people on the other side.

2 RELATED WORK

Illinois Experts [1] and Google Scholar search researchers by keyword but ignore network position: your co-author's co-author ranks like a stranger. LinkedIn's "People You May Know" and Twitter's Who-to-Follow [6] use triadic closure, the opposite of brokerage. Link prediction [7] finds *likely* ties; we rank ties by *value*, using Burt's constraint [3].

3 COURSE CONCEPTS & NETWORK ANALYSIS

Core concept: **structural holes and brokerage** (Lesson 1), with **community discovery** (Lesson 10) and **weak ties and decentralized search** (Lessons 1, 13).

Graph and method. Nodes are UIUC researchers in OpenAlex [9]. Edges are 2015–2024 co-authorships weighted by joint papers. Each node has a topic vector. (1) Louvain modularity [2] finds communities for the map. (2) Two communities form a *hole* when their topics are similar but few edges join them. (3) For a user u , each researcher v is scored by $\text{topic similarity} \times \text{graph distance} \times \text{brokerage gain}$, the drop in u 's constraint if the tie $u-v$ were added. (4) A weak-tie shortest path offers an introduction.

Alternatives from class. Triadic closure only finds people already near you; topic similarity alone can't tell a stranger from a friend of a friend. Girvan–Newman [8] could not split a 150-node graph in 148 s in our pilot; Louvain split 610 nodes in 1.5 s.

Claim that could be false. From the 2015–19 graph, BridgeFinder ranks the actual partners of new *cross-community* collaborations

(2020–24) higher than triadic-closure or topic-only ranking, and those collaborations' papers are cited more than new within-community ones in the same fields (Burt's claim).

4 PROPOSED WORK

A web app for Figure 1 (Python/networkx over an OpenAlex snapshot) plus the backtest. **Oct 30:** graph, holes, backtest, prototype, feedback round 1. **Mid-Nov.:** person cards, intro paths, round 2. **Dec 9:** round 3, evaluation. *Assumptions:* OpenAlex disambiguation is good enough (we will hand-check 50 authors); topic vectors capture "similar research"; students without papers are placed through their advisor or lab; using only public publication data keeps suggestions appropriate.

5 PRELIMINARY DESIGN

In Figure 1, the map comes first, so users see *where* the holes are before seeing names. Every suggestion shows its reasons, because unexplained recommendations are hard to trust.

6 USERS

UIUC students who already do research (a paper or an advisor, so they have a place in the graph) and want mentors or collaborators outside their group. We will recruit about 10 from our networks and classes and reuse them across three rounds. Ratings are blind, so friendliness cannot favor our method.

7 EVALUATION

(1) **Backtest (no users).** With 2015–19 data and only researchers ≥ 3 steps from each user (whom we recommend), how often do users' real 2020–24 new co-authors land in our top 10 vs. topic similarity alone? We expect triadic closure to win on *all* new ties and will report it. (2) **Blind rating.** Participants rate a mixed, unlabeled list from ours, triadic closure and topic-only as relevant (1–5) and new or not; we succeed if more of ours are both. (3) **Usefulness.** Would they feel comfortable contacting each person? (No email is sent.) *Use of AI:* Claude helped brainstorm and draft; we verified everything.

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